Vataleni PMSG SSO Validation Report

Fresh execution record for the saved Simulink model. The test suite completed on 21 October 2026 in MATLAB R2025b and Simulink 25.2.

Saved SLX SHA-256: `d4dc9bcc27fa0a904b43eb40914887ab14503bafc3769898805618f43cdb4aaa`

Execution result

The 20-check project validator reported **20/20 PASS** against the saved SLX. It opened and compiled the model without a project initializer, completed the 12 s baseline, weak-grid SSO and mitigated scenarios, and regenerated 17 figures and numerical tables. The model has zero external custom S-functions and zero unresolved links. All 12 native Scopes produced finite logged Dataset outputs in a separate saved-model run.

Check	Status	Evidence
Model opens	PASS	load_system completed from repository root without project paths
Model compile/update	PASS	SimulationCommand update completed
No unresolved custom S-functions	PASS	External/custom S-Functions: 0; embedded MATLAB Function engines: 1
No broken library links	PASS	Unresolved links: 0
Configured simulation completes	PASS	60001 time samples to t=12.000 s
All outputs finite	PASS	Finite values: 2100035 of 2100035
Baseline scenario	PASS	P_end=0.881502 pu; max Vdc deviation=0 pu
SSO scenario	PASS	FFT=4.84467 Hz; eigenvalue=0.047491+30.414487i
Mitigation scenario	PASS	Power oscillation RMS 0.0677542 -> 2.65677e-07 pu
Figures regenerate	PASS	17 fresh PNG figures generated
Numerical result exports regenerate	PASS	Required exports present: 5/5; dt 2.0e-04 -> 1.0e-04 s, frequency change 0 Hz

Scenario metrics

All values below are from the reduced-order model, not copied from a literature plot. FFT is computed from the grid active-power response; damping comes from the linearized model and the documented signal analysis.

Case	SCR	FFT Hz	Power oscillation RMS pu	Peak current pu
sso_before	5	4.8447	0.0677542	0.9744
sso_after	5	1.8883	2.65677e-07	0.9144

Numerical checks

The baseline logged 60,001 samples over 12 s and 35 channels, all finite. A second MATLAB process opened a copy of the exact SLX from a folder containing no project files, compiled it and completed another 60,001-sample baseline run with native Scope data. The study step was 0.0002 s; repeating the weak-grid frequency extraction at 0.0001 s produced the same reported FFT bin. The project validator reported no algebraic loops, NaN, Inf, unresolved links or machine-specific paths.

Independent engineering checks

Mechanical power and PCC active/reactive power residuals were zero to the recorded precision in all four study cases. The capacitor energy-balance residuals below were calculated from the saved model outputs and a finite-difference derivative; their nonzero values are dominated by numerical differentiation at transients.

Case	DC energy residual RMS pu	Max DC voltage deviation pu
baseline	0	0
wind_step	1.17e-10	8.194e-05
sso	3.58e-07	0.04487
mitigation	8.24e-08	0.01485

Interpretation limits

Passing the existing validator establishes internal reproducibility of this averaged benchmark. The validator's trajectory comparison uses the project's recovered monolithic equations as its reference, so it is not an independent physical validation. The current model does not satisfy the requested full PMSG stator d/q dynamics or native Simscape Electrical component representation. It also retains optional historical MATLAB analysis scripts in the repository. These acceptance items remain open and should not be marked PASS.

Reproduction

Open model/Vataleni_PMSG_SSO_Final.slx in a fresh MATLAB session and press Run for the baseline. Set Model Workspace values as listed in the Model Guide for the two weak-grid cases. The complete validator is validation/validate_project.m; it is optional external verification tooling, not required by the SLX.